

Datathon 2026

GLOBAL HEALTHCARE ANALYSIS

TEAM: FINTECH JUNKIES

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AGENDA

- **OUR PROBLEM**
- **INITIAL ANALYSIS**
- **MACHINE MODEL**
- **OUR SOLUTION**

PROBLEM

- **LIMITED HEALTH RESOURCES**
- **POSSIBLE
UNDERINVESTMENT OR
OVERINVESTMENT IN
CERTAIN COUNTRIES**

GOAL

**IDENTIFY WHERE A FIXED BUDGET
CAN REDUCE HEALTHCARE SYSTEM
STRAIN THE MOST PER DOLLAR**



DATASET

- 20 COUNTRIES
ACROSS EVERY
MAJOR CONTINENT
- 2000-2024
- DISEASE AND YEAR
ISSUE IN DATA

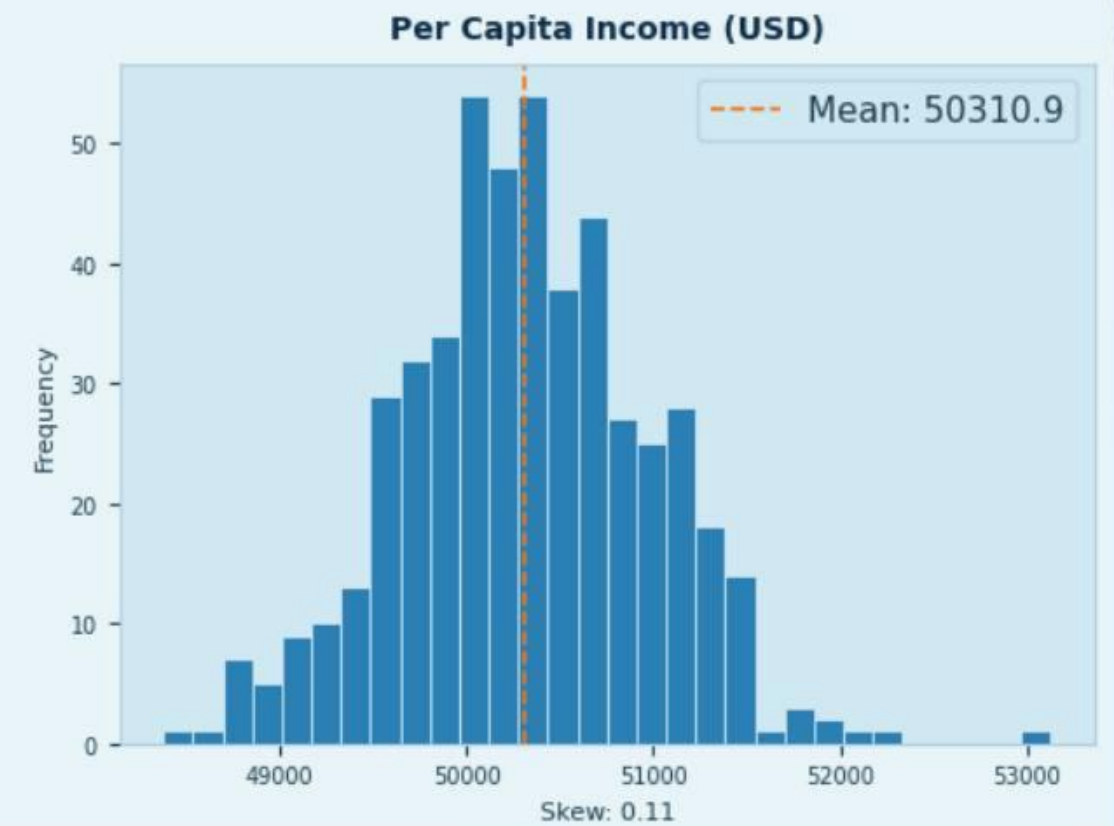
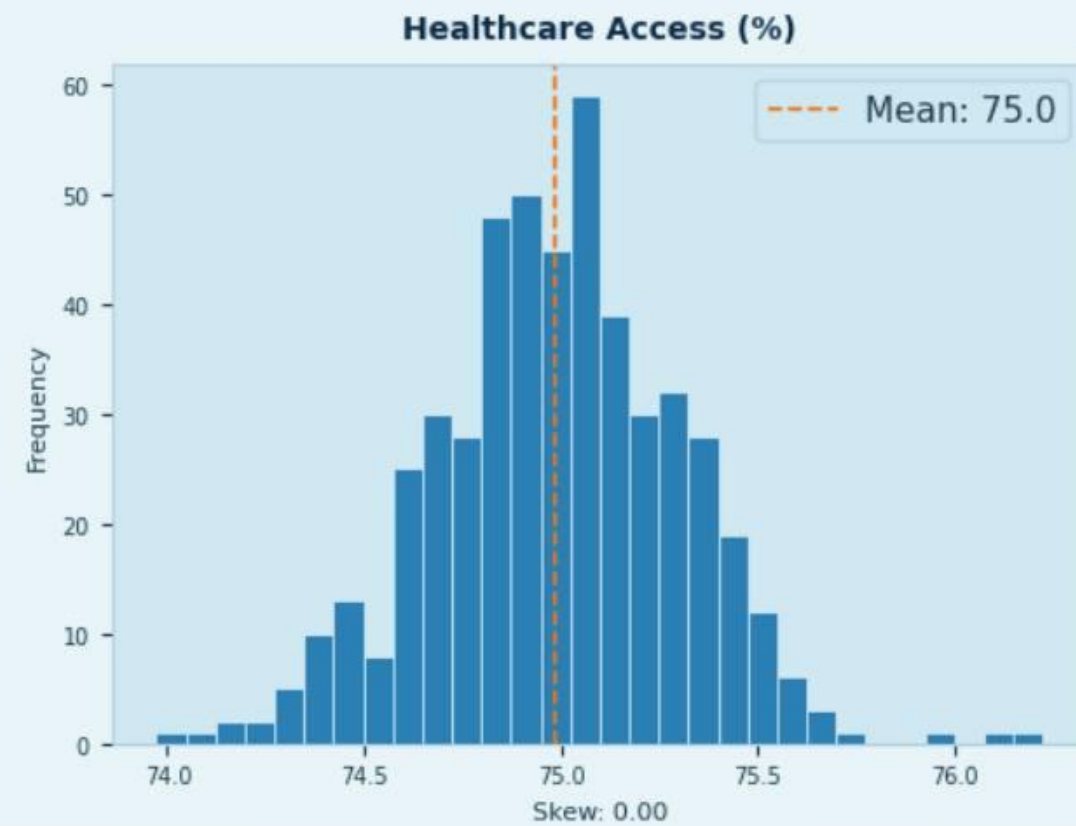
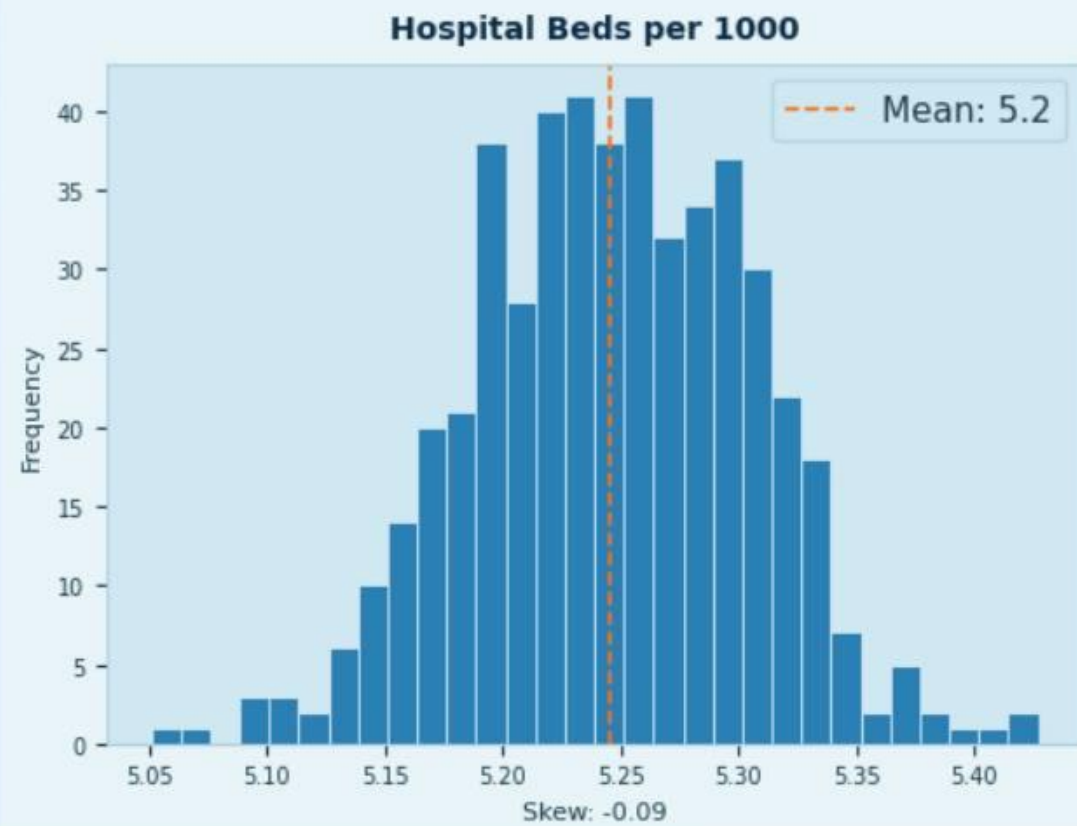
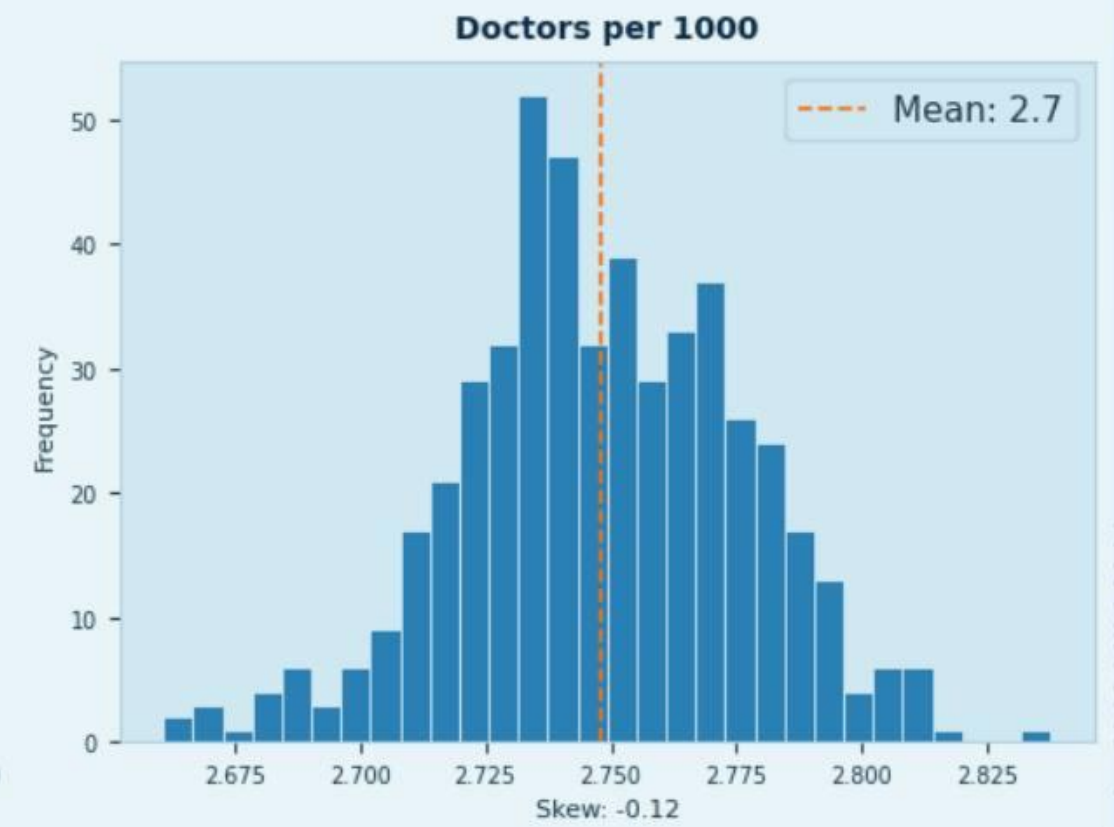
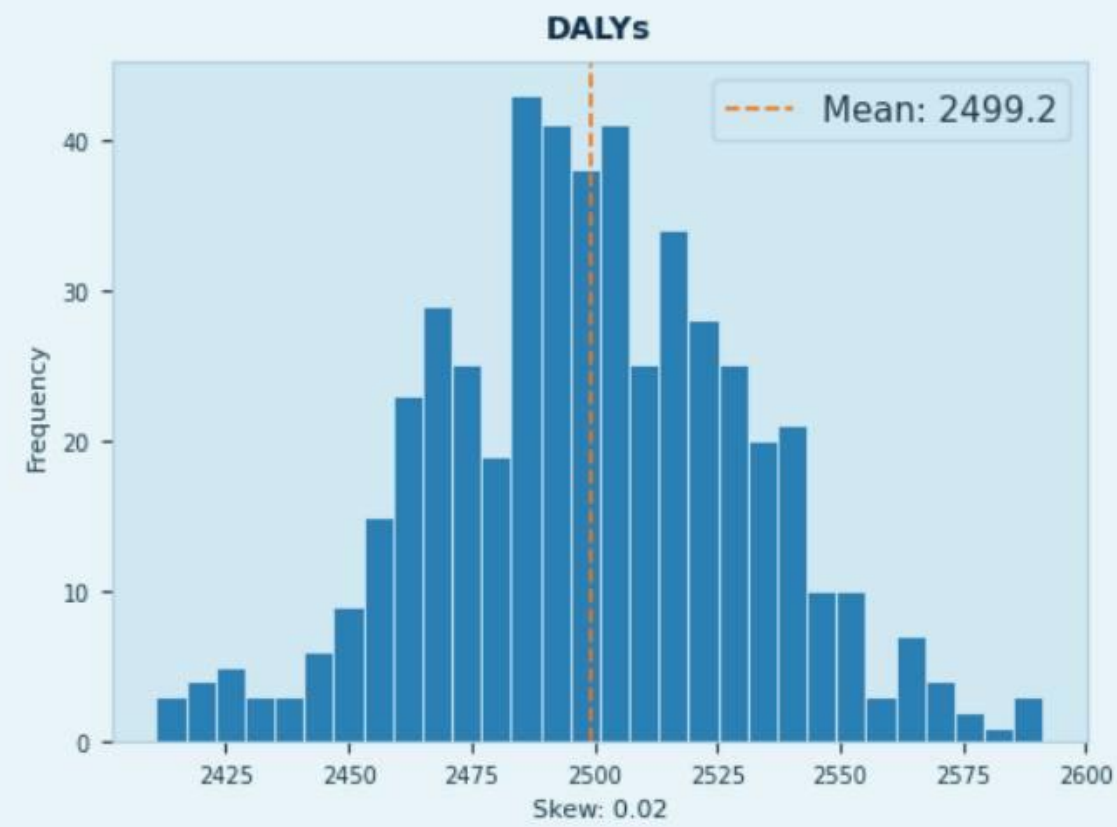
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['Country',  
'Year',  
'Disease Name',  
'Disease Category',  
'Prevalence Rate (%)',  
'Incidence Rate (%)',  
'Mortality Rate (%)',  
'Age Group',  
'Gender',  
'Population Affected',  
'Healthcare Access (%)',  
'Doctors per 1000',  
'Hospital Beds per 1000',  
'Treatment Type',  
'Average Treatment Cost (USD)',  
'Availability of Vaccines/Treatment',  
'Recovery Rate (%)',  
'DALYs',  
'Improvement in 5 Years (%)',  
'Per Capita Income (USD)',  
'Education Index',  
'Urbanization Rate (%)']
```

STRAIN INDEX

- **MEASURED HEALTHCARE BURDEN ACROSS 20 COUNTRIES**
- **AGGREGATED 4 DISEASE BURDEN SIGNALS PER COUNTRY**
 - **RECOVERY RATE**
 - **IMPROVEMENT OVER 5 YEARS**
 - **MORTALITY RATE**
 - **DALYS (DISABILITY-ADJUSTED LIFE YEARS)**

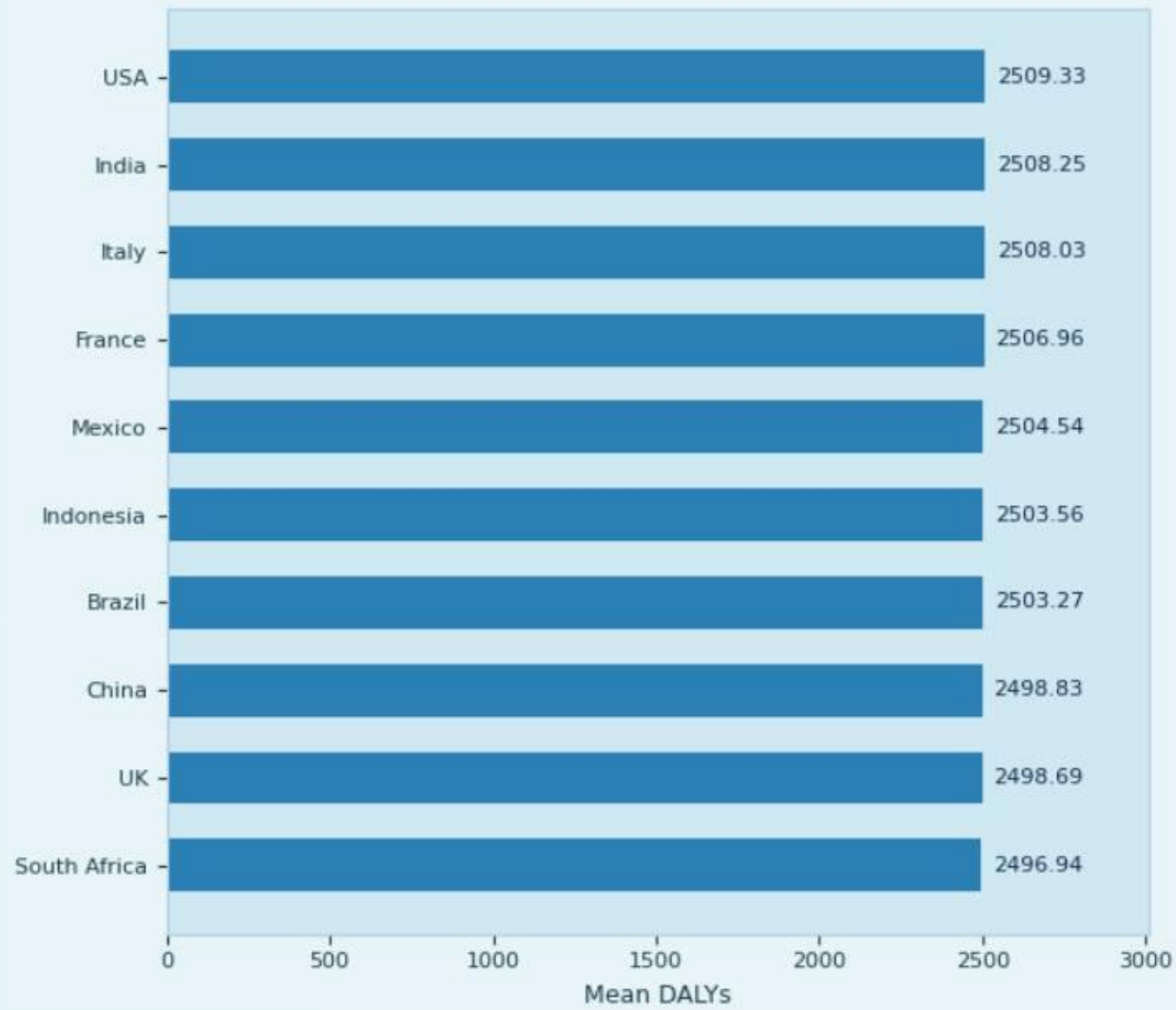
INITIAL EDA

Feature Distributions



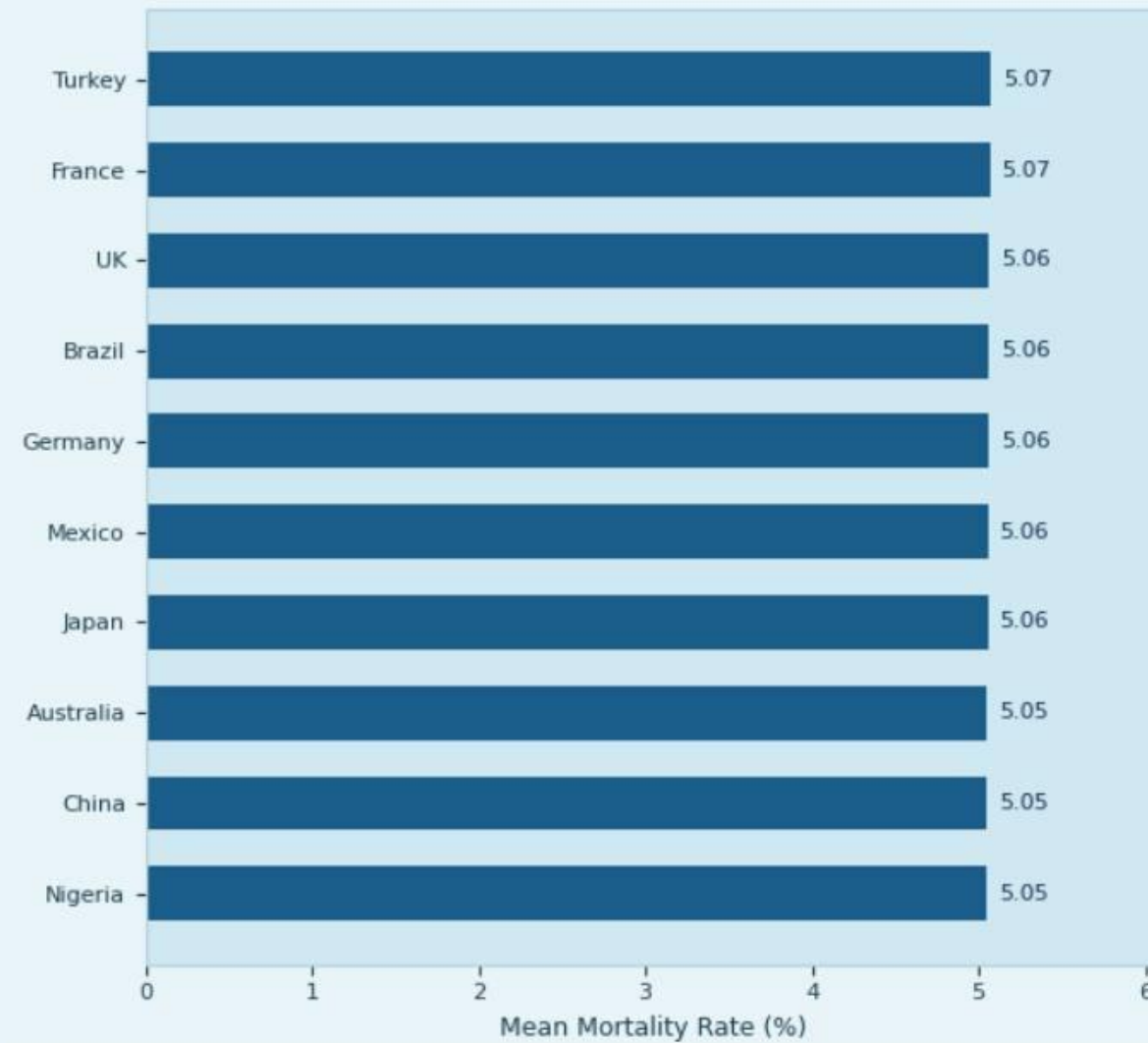
INITIAL EDA

By Mean DALYs

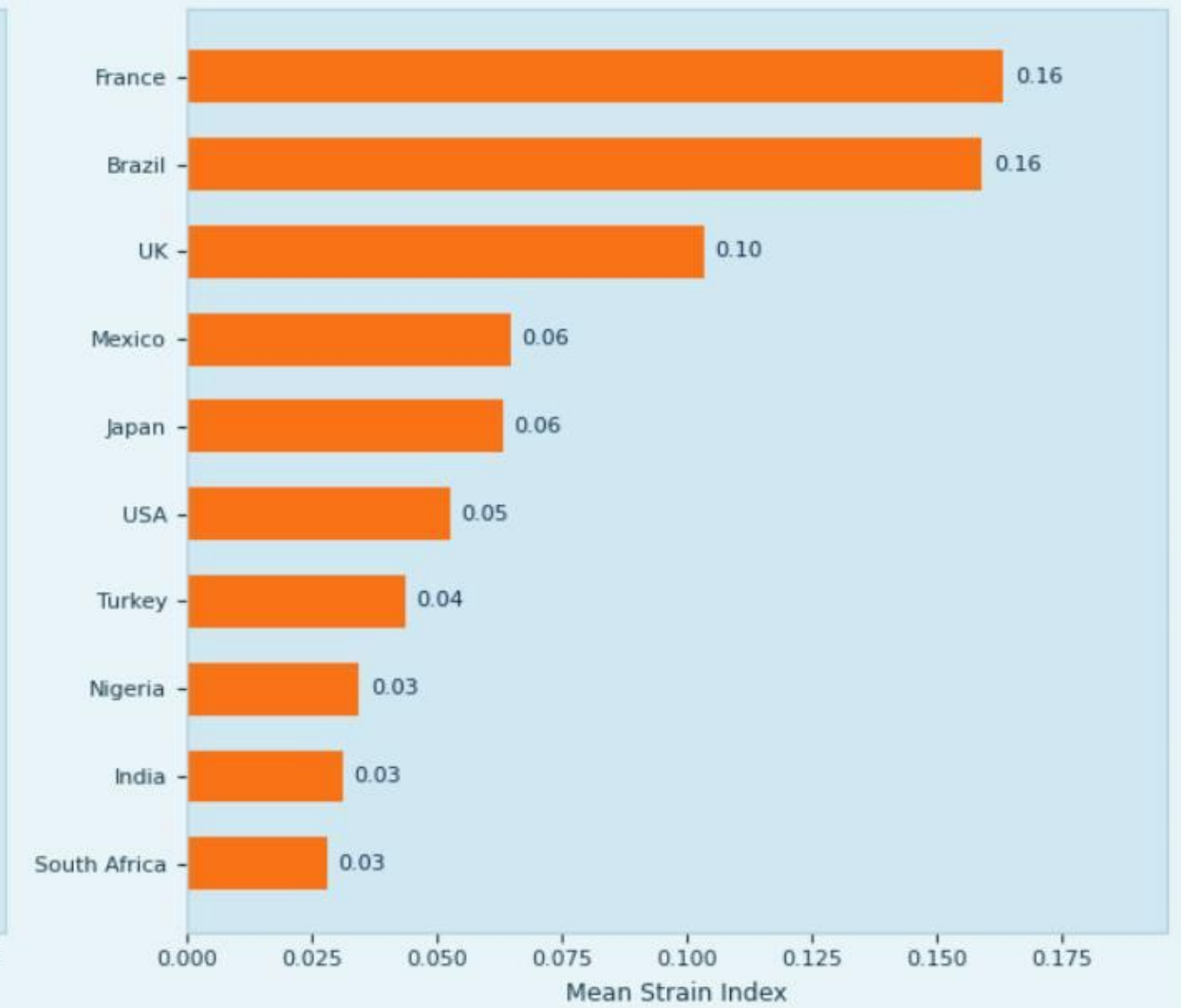


Top 10 Most Strained Countries

By Mean Mortality Rate (%)

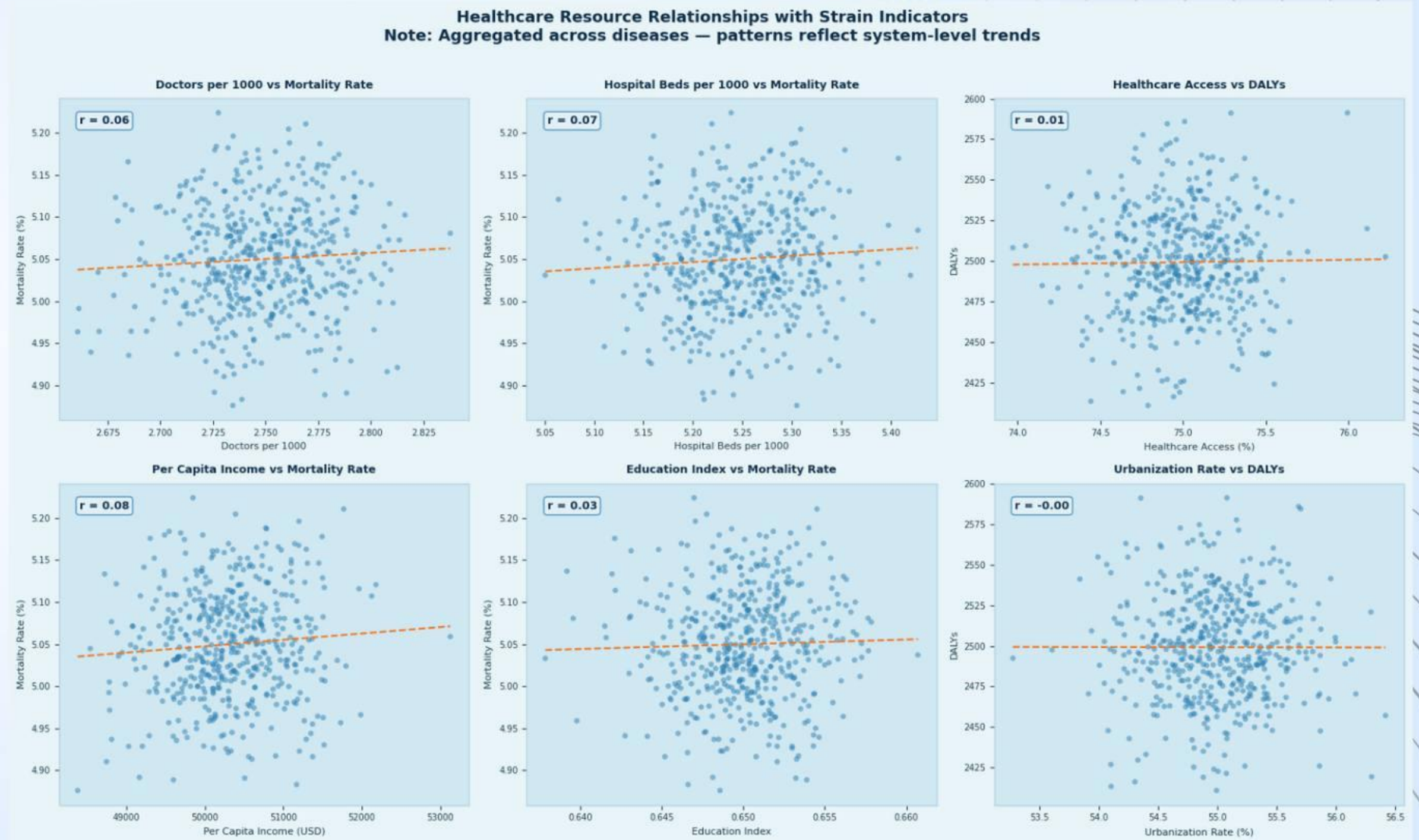


By Mean Strain Index

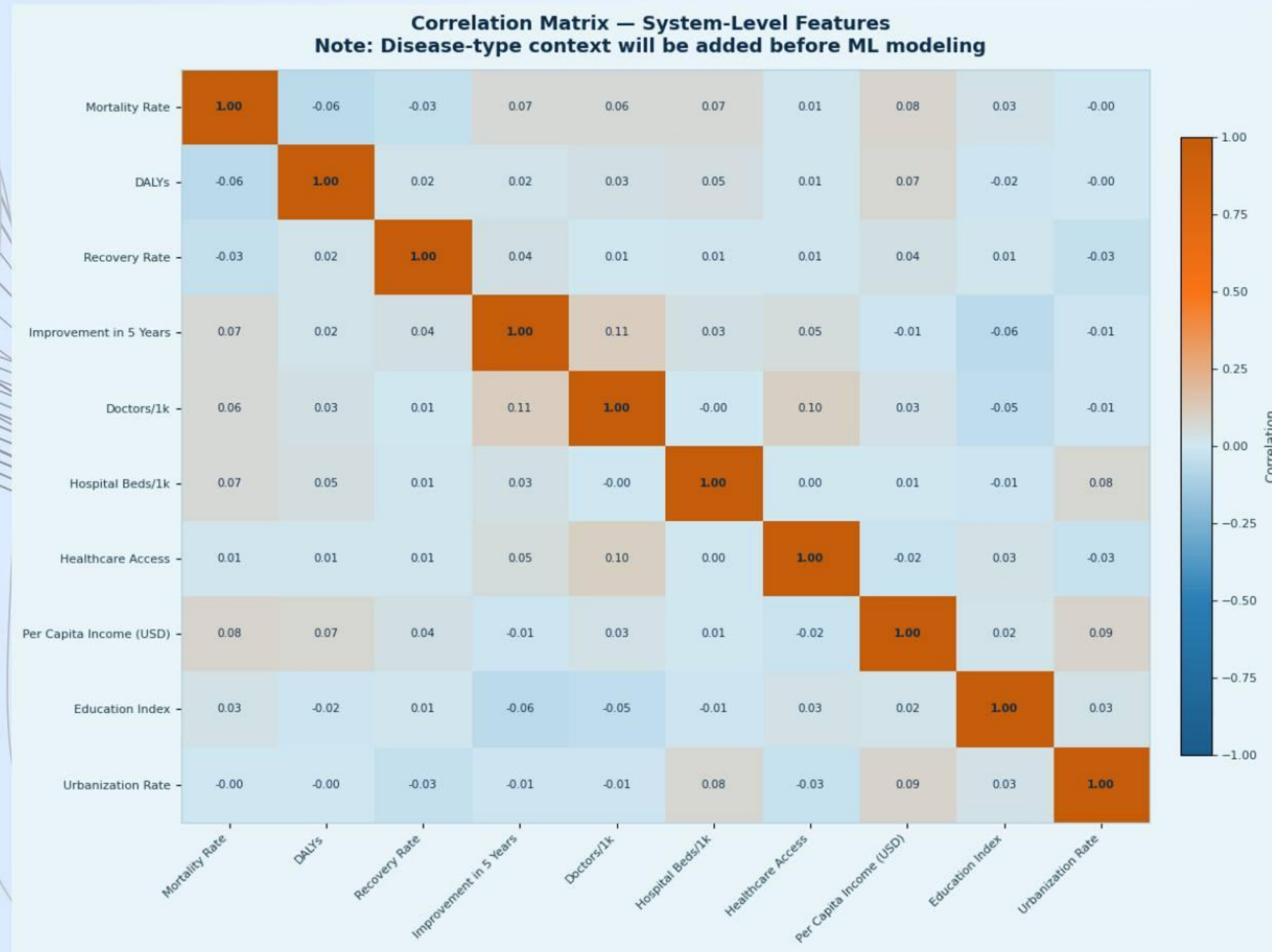


CORRELATION

- COMPARED RELATIONSHIP OF FEATURES VS BURDENS
- NO MAJOR NOTICEABLE CORRELATION BASED OFF OF SCATTERPLOTS



CORRELATION



- FEATURES RELATIONSHIP
- “STRONGEST” FEATURE RELATIONSHIP BETWEEN DOCTORS AND IMPROVEMENT IN 5 YEARS

STRAIN INDEX 2.0

- **BURDEN ALONE ISN'T ENOUGH** - A SICK COUNTRY WITH STRONG HOSPITALS ISN'T STRAINED
- **THREE PILLARS:**
 - **BURDEN SCORE** - RECOVERY, IMPROVEMENT, MORTALITY, DALYS
 - **CAPACITY GAP** - SHORTAGE OF DOCTORS, BEDS, AND HEALTHCARE ACCESS
 - **AFFORDABILITY GAP** - TREATMENT COST RELATIVE TO INCOME

$$\text{STRAININDEX} = (0.5 \times \text{BURDENSORE}) + (0.3 \times \text{CAPACITYGAP}) + (0.2 \times \text{AFFORDABILITY GAP})$$

MACHINE MODEL

MODEL SETUP

- RANDOM FOREST REGRESSOR
- 400 DECISION TREES
- TRAIN: YEARS $\leq$ 2022 TEST: YEARS $\geq$ 2023
- TARGET: STRAIN INDEX

FEATURES

- HEALTHCARE ACCESS
- DOCTORS PER 1,000
- HOSPITAL BEDS PER 1,000
- PER CAPITA INCOME
- COST-TO-INCOME RATIO

- **BURDEN SIGNALS INTENTIONALLY EXCLUDED** - MODEL ONLY HAS INPUTS WE CAN BUY WITH MONEY
- **OUTPUTS** - PREDICTS STRAIN & BEST ACTION PER COUNTRY + ROI

\$100M ALLOCATION PLAN

HOW IT WORKS

- GREEDY OPTIMIZER: LOOPS AND EACH ITERATION ALWAYS PICKS HIGHEST ROI ACTION LEVER
- $ROI = \text{STRAIN REDUCTION} / \text{COST}$
- COSTS SCALED BY COUNTRY INCOME LEVEL
- REPEATS UNTIL BUDGET EXHAUSTED

RESULTS

- 81 INVESTMENT DECISIONS
- \$99.69M DEPLOYED | \$310K REMAINING
- 3 INTERVENTION TYPES: ACCESS (\$1M BASE), DOCTORS (\$1.5M BASE), BEDS (\$1.2M BASE)

TOP RECIPIENTS

- SOUTH AFRICA - LARGEST IMPROVEMENT (-0.104)
- ARGENTINA
- TURKET
- JAPAN, UK ALSO RECEIVED MULTIPLE ROUNDS

RESULT: EVERY INVESTMENT WENT TO THE COUNTRY + ACTION LEVER WITH THE HIGHEST ROI AT THAT MOMENT

CONCLUSIONS

- **STRAIN INDEX 2.0 IDENTIFIED ITALY, SAUDI ARABIA, SOUTH KOREA AS HIGHEST STRAIN IN 2024**
- **\$100M DEPLOYED OPTIMALLY ACROSS 81 STEPS - SOUTH AFRICA BENEFITED THE MOST (-0.104)**
- **BEST ACTION LEVER INVESTMENT IS COUNTRY-SPECIFIC: ACCESS FOR BRAZIL/SAUDI ARABIA, BEDS FOR ITALY, DOCTORS FOR JAPAN**

SOLUTION

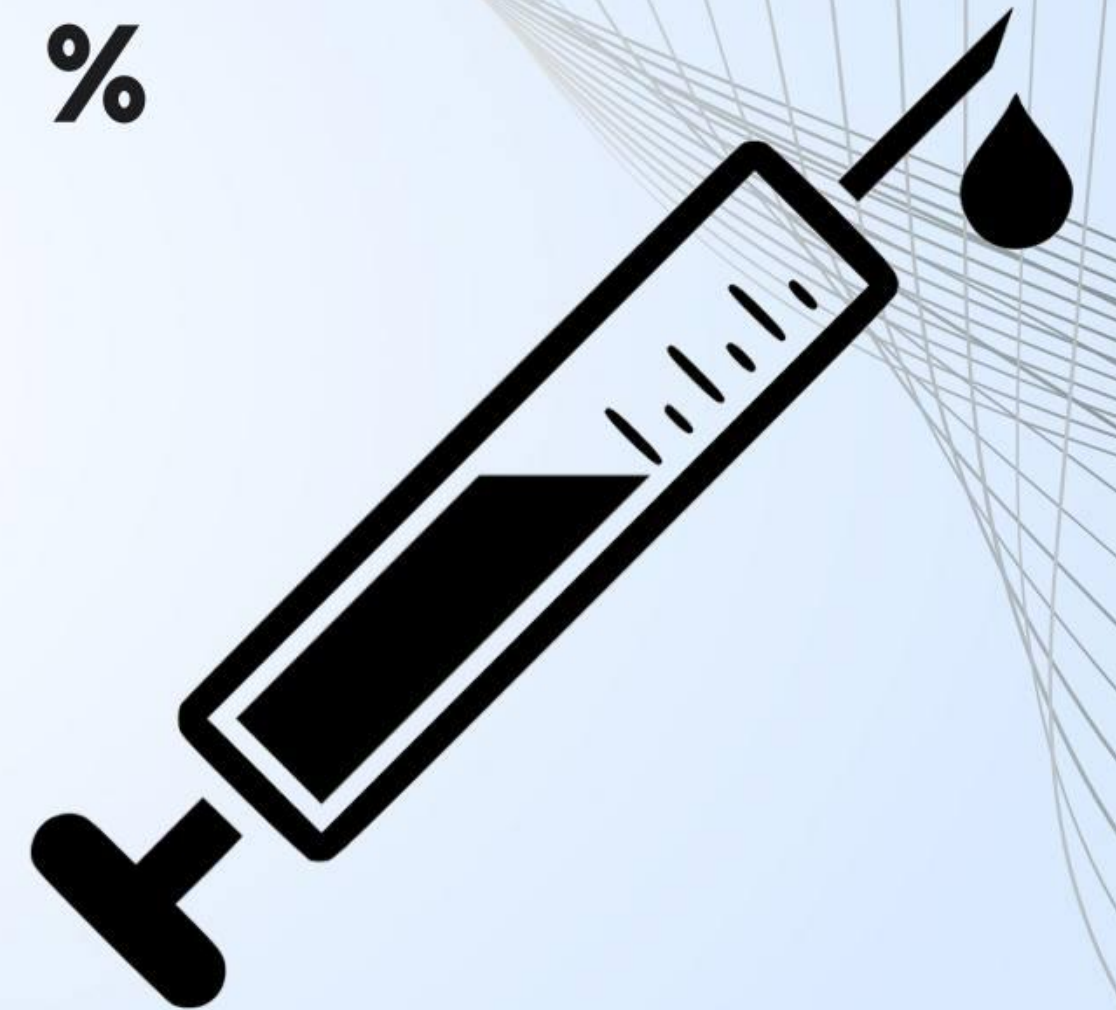
**INVEST IN:
ACCESS
WORKFORCE
CAPACITY**

HOW?



ACCESS

- **EFFECT: HIGHER ACCESS %**
- **VACCINATIONS**
- **DISEASE SCREENING**
- **PRIMARY CARE**
- **SUBSIDIZE COSTS**
- **\$1M PER .5% INCREASE**



WORKFORCE EXPANSION

- **EFFECT: INCREASE DOCTORS PER 1000**
- **FUND MEDICAL SCHOOLS**
- **TRAINING PROGRAMS**
- **\$1.5M PER .01 IN 1000**



CAPACITY (BEDS)

- **EFFECT: HIGHER BEDS
PER 1000**
- **CONSTRUCT MORE
HOSPITALS**
- **EXPAND CURRENT
HOSPITALS**
- **\$1.2M PER .02 IN 1000**





QUESTIONS?